



## First prosthetic that can feel

Researchers have developed a prosthetic that can provide a form of feedback to the wearer based on touch <http://dgit.in/FeelHand>

absence of gravity, these traps can be made weaker, allowing the gases to expand and cool more than it would have under the influence of gravity.

## The Blackest

The blackest black right now is debatably, a black hole. It sucks all light and obviously reflects none, but even that is debatable. In 2008, scientists contributed to this world the blackest man-made object yet – the VACNT.

By aligning carbon nanotubes vertically in a brush-like formation, Rennselaer Polytechnic Institute created an object which absorbs 99.97% of all light. A light photon which does not get absorbed by one nanotube gets deflected and absorbed by another, a process which repeats until the light is completely absorbed by what the researchers have called the 'forest'. Due to its nature, The VACNT with the naked eye will appear perfectly flat when in fact it is a sheet full of deep holes.

Considering that conventional black coatings or graphite generally reflect about 5% of light, 99.97% absorption is pretty neat. Especially when even NASA was not able to beat it, creating a substance which absorbs only what is now a laughable 99.5%. According to fastcompany.com though, NASA, in collaboration with Melbourne Center for Nanofabrication, have recently worked on a project to create something even blacker.

The burning question is of course why? What need do we have for this abomination? Well since it absorbs almost all light, it would be a very useful in energy converters and a well as in stealth and defence as it could serve as a brilliant cover. Splinter Cell would get a bit easier, among other things.

## The Lightest

Okay so here is a category which gets reshuffled very regularly. A type of Aerogel or "frozen smoke" with a density of 4 mg/cm<sup>3</sup> came into prominence for being the lightest material in the world in 2008. In 2011, good old frozen smoke lost its title to a micro-lattice material with a density of 0.9 mg/cm<sup>3</sup>. Aerographite soon followed with a density of 0.18 mg/cm<sup>3</sup>.

The current titleholder, beating the previous contestant by a sweet 11% in weight, is the Graphene Aerogel, created by researchers at Zheijang University in China, with a density of 0.16 mg/cm<sup>3</sup>. Created with the magical material Graphene, the researchers have claimed that if our entire body was made up of this material, we would weigh only around 0.01kg.

The lightest man-made material in the world (for now) was created by taking a semi-solid gel of carbon nanotubes and graphene and then freeze-drying it to convert it to a sponge-like material which can be adjusted into whatever shape desired. This process, the researchers said, is actually one of the easiest method in producing a light material, thus kicking the behinds of previous materials in more than one possible area.

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